Problem Set 6

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Consider a one-dimensional data set (x_n, y_n), $n = 1, \ldots, N$, and ridge regression with the intercept fixed at zero:

$$L(w) = \frac{1}{2}\sum_{n=1}^{N}(y_n - x_n w)^2 + \frac{\lambda}{2}\,w^2, \qquad \lambda \geq 0.$$

(a) Derive, step by step, a closed-form expression for the minimizer $w^\star$. Show that it is a minimizer and not just a stationary point.

(b) Intuitively, what effect does increasing λ have on the bias of $w^\star$? On its variance?

Problem 2

Suppose $n = 2$ and $d = 2$, with

$$x_{11} = x_{12} = a, \qquad x_{21} = x_{22} = -a \quad (a \neq 0), \qquad y_1 + y_2 = 0.$$

Because the responses and each feature column sum to zero, the intercept estimate is zero and can be dropped. The ridge loss is

$$L(w) = \frac{1}{2}\sum_{i=1}^{2}\left(y_i - x_{i1}w_1 - x_{i2}w_2\right)^2 + \frac{\lambda}{2}\left(w_1^2 + w_2^2\right), \qquad \lambda > 0.$$

(a) Show that in this setting

$$L(w) = \big(y_1 - a(w_1 + w_2)\big)^2 + \frac{\lambda}{2}\left(w_1^2 + w_2^2\right).$$

(b) Show that the ridge coefficients satisfy $w_1^\star = w_2^\star$, and find their common value.

(c) Compute $w^\star$ for $a = 1$, $y_1 = 2$, $\lambda = 1$.

Problem 3

Let

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}, \qquad A + I = \begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix}.$$

(a) Find the eigenvalues and eigenvectors of A.

(b) Find the eigenvalues and eigenvectors of $A + I$. How are they related to those of A?

(c) What does one of the eigenvalues of $A + I$ tell you about the matrix $A + I$?

Problem 4

A symmetric matrix acts as a pure scaling when points are described in coordinates along its eigenvectors. Every symmetric $A \in \mathbb{R}^{2\times 2}$ can be written $A = Q\Lambda Q^T$, where the columns q_1, q_2 of Q are orthonormal eigenvectors and $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2)$. Reading $Ax = Q\Lambda Q^T x$ from right to left gives three steps:

(i) $u = Q^T x = (q_1^T x,\ q_2^T x)^T$: the coordinates of x along the eigenvectors;

(ii) $v = \Lambda u$: scale the two coordinates by λ_1 and λ_2;

(iii) $y = Qv = v_1 q_1 + v_2 q_2$: convert back to standard coordinates.

Use

$$A_1 = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} = \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}}_{\Lambda} \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q^T},$$

$$A_2 = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} = \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}}_{\Lambda} \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}_{Q^T},$$

and the points $x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $x_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $x_3 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Note that both matrices share the same Q.

(a) Write down q_1, q_2, λ_1, λ_2 for each matrix. Check that $Q^TQ = I$ and compute $\det Q$: is Q a rotation or a reflection?

(b) Compute A_1x and A_2x directly for each of the three points.

(c) Compute $u = Q^Tx$ for each point (only once, since Q is shared). Which u has a zero entry, and why? Check that $\|u\| = \|x\|$; what does this say about how Q^T treats lengths?

(d) Compute $v = \Lambda u$ for each matrix. Along each eigenvector direction, is the point stretched or shrunk, by what factor, and is it flipped?

(e) Compute $y = Qv$. Record x, u, v, y in a table for each matrix, and confirm that each y matches part (b).

(f) Interpret your results.

(i) Which point is only scaled, never turned, by both matrices? Why?

(ii) Using the eigenvalues, explain why A_1 pulls x_1 and x_3 towards the diagonal.

(iii) Show that $A_2 = 2\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and describe what A_2 does to the plane.

(g) Sketch the image of the unit circle under A_1 and under A_2, and relate its semi-axes to the eigenvectors and eigenvalues. Why is the flip in A_2 invisible in the shape of the image? *Hint: track where x_1 and x_3 go.*

Problem 5

This problem uses the eigenvalue decomposition to compute powers of a matrix, and links eigenvalues to the characteristic polynomial.

(a) Diagonalize

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

and compute $Q\Lambda^kQ^{-1}$ to prove that

$$A^k = \frac{1}{2}\begin{pmatrix} 1+3^k & 1-3^k \\ 1-3^k & 1+3^k \end{pmatrix}.$$

(b) If a 2×2 matrix B has eigenvalues $\lambda_1 = 3$ and $\lambda_2 = 4$, what is $\det(B - \lambda I)$?